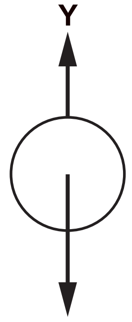
Examiner only

**5.** A ball is dropped from rest.
Two forces, X and Y, act on the ball during its fall.




(a) (i) Name the forces X and Y. [2]

X ..................................................................... Y .....................................................................

(ii) State what happens to the size of the forces, if anything, as the ball speeds up. [2]

Force X ..................................................................................................................................................

Force Y ..................................................................................................................................................

(b) Complete the following sentences by underlining the correct phrase in the brackets. [2]

(i) The ball speeds up at the start of its fall because force Y is (**smaller than** / **equal to** / **larger than**) force X.

(ii) The ball eventually reaches a constant speed because force Y is (**smaller than** / **equal to** / **larger than**) force X.

(c) After being dropped from rest, the ball takes 2 seconds to hit the ground.

(i) Use the equation:

$$v = u + at$$

to calculate the final speed, $v$, with which the ball hits the ground. [2]
(Acceleration due to gravity, $g$ = 10 m/s$^2$)

$v$ = ................................ m/s

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(ii) Use an equation from page 2 and your answer to part (c)(i) to calculate the height, $x$, from which the ball was dropped. [3]

$x$ = ................................ m

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